Fashion Lens

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Abstract

Fashion e-commerce platforms require accurate product categorization to enhance searchability and user experience. Manual tagging is inefficient, inconsistent, and time-consuming, necessitating an automated solution. Fashion Lens is an AI-powered image classification system that leverages deep learning, specifically Convolutional Neural Networks (CNNs), to classify fashion products with over 90% accuracy. The system processes clothing images, extracts visual features, and assigns predefined labels, reducing manual effort and improving operational efficiency. Integrated via an API-based approach, Fashion Lens enables real-time classification, making it scalable for large e-commerce platforms. This innovation enhances search results, recommendation systems, and product cataloging, paving the way for advanced automation in fashion retail. Future improvements include multi-category tagging, fabric recognition, and augmented reality (AR) integration, further revolutionizing the online shopping experience.

Introduction

Fashion e-commerce platforms rely on accurate product categorization to enhance user experience, improve searchability, and optimize inventory management. Traditional manual tagging is inefficient, inconsistent, and time-consuming, making automation essential. Fashion Lens is an AI-powered image classification system that leverages deep learning, specifically Convolutional Neural Networks (CNNs), to automatically categorize fashion products with high accuracy. The system processes clothing images, extracts key visual features, and assigns predefined labels, reducing manual effort and improving scalability. Integrated with an API-based architecture, Fashion Lens enables real-time classification, making it suitable for large-scale e-commerce platforms. By enhancing automation, minimizing operational costs, and improving recommendation systems, this solution transforms online shopping experiences. Future advancements, such as multi-category tagging, fabric recognition, and trend prediction, will further revolutionize fashion retail through AI-driven innovation.

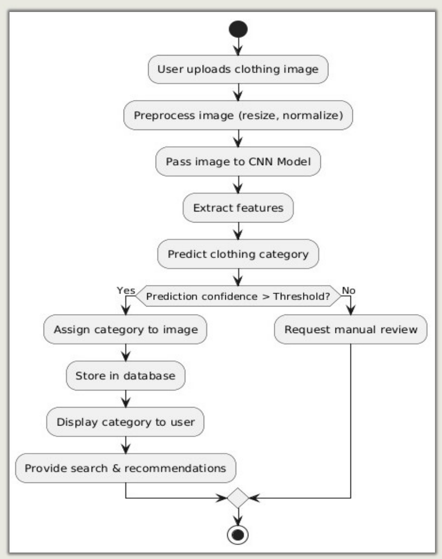
Objectives

The primary goal of Fashion Lens is to automate the process of fashion product categorization using deep learning techniques, reducing reliance on manual tagging while significantly improving efficiency, accuracy, and scalability. Traditional manual classification is often inconsistent, time-consuming, and labor-intensive, making it impractical for large-scale e-commerce platforms. By leveraging Convolutional Neural Networks (CNNs), this system enhances searchability, product recommendations, and inventory management, ultimately improving the overall shopping experience for users. Additionally, the project aims to provide an API-based integration that allows real-time classification, making it easily adaptable for online retail platforms.

Methodology

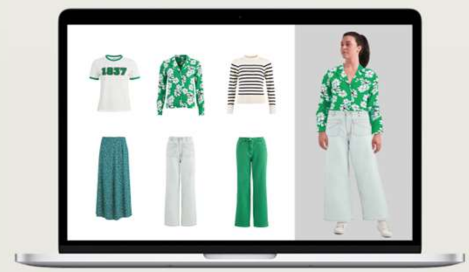
The Fashion Lens system follows a structured methodology that includes data collection, preprocessing, model training, evaluation, and deployment:

- ✓ Data Collection & Preprocessing: A diverse dataset of fashion products was gathered, consisting of various categories such as shirts, dresses, pants, and accessories. Images were resized, normalized, and augmented using techniques like flipping, rotation, and brightness adjustments to improve model robustness and generalization.
- ✓ Model Selection & Training: A ResNet-50-based CNN model was fine-tuned using transfer learning to extract relevant visual features from fashion products. The dataset was split into training, validation, and testing sets, and the model was optimized using the Adam optimizer with a learning rate scheduler.
- ✓ Evaluation Metrics: The model's performance was assessed using accuracy, precision, recall, F1-score, and a confusion matrix to ensure its reliability.
- ✓ Deployment: The trained model was integrated into an API-based system using Flask and FastAPI, allowing real-time classification. The API processes images and assigns predefined category labels within milliseconds, making it highly suitable for large-scale e-commerce platforms.



Results

The model achieved an accuracy of over 90%, significantly outperforming traditional rule-based classification methods. The deep learning approach enabled more consistent and efficient product categorization, reducing human errors and improving classification speed. Performance evaluations demonstrated high precision and recall, ensuring that products were correctly labeled with minimal misclassification. Additionally, the system's real-time processing capabilities made it a scalable solution for e-commerce businesses, allowing seamless integration into their existing infrastructure.



Conclusion

Fashion Lens presents a cutting-edge AI-powered solution for fashion product categorization, addressing the inefficiencies of manual tagging and revolutionizing inventory management for e-commerce platforms. By leveraging deep learning, automation, and real-time classification, the system enhances searchability, recommendation accuracy, and operational efficiency, ultimately improving the online shopping experience. The findings of this study highlight the potential of AI in transforming retail, reducing operational costs, and enabling faster product discovery. Future advancements, such as multi-category tagging, fabric recognition, trend prediction, and augmented reality (AR) integration, will further expand the system's capabilities, making AI-driven fashion retail more dynamic and intelligent.

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